Seeing the Curve

Summary

PCA on daily U.S. Treasury yield changes identifies the level, slope, and curvature factors. A PC1/PC2-neutral butterfly built from these loadings cleanly isolates curvature. A z-score mean-reversion strategy on curvature generates near-zero gross profit and incurs transaction costs that dwarf any signal edge.

Data Pull & Fixes

Daily constant-maturity yields for 2Y, 5Y, 7Y, 10Y, 20Y, and 30Y tenors from January 2019 to December 2025 were pulled via OpenBB. The cleaning pipeline separates detection from removal with five checks covering non-numeric values, NaNs, missing business dates, non-business-day rows, and duplicates. All checks came back clean for this data pull.

PCA & Factor Stability

PCA was run on the covariance matrix of unstandardized daily yield changes because all tenors share the same unit and cross-tenor volatility differences are economically meaningful information needed for DV01-based hedging. Three PCs explain 99.3% of variance with individual shares of 88.3%, 9.7%, and 1.3%. PC1 is the level factor with all loadings positive, PC2 is the slope factor with a monotonic sign flip from +0.68 at 2Y to −0.50 at 30Y, and PC3 is the curvature factor with wings positive and belly negative.

To assess factor stability, the sample was split into a pre-hike regime and a post-hike regime starting from Mar 16, 2022, with the cut chosen at the first Fed rate hike of the tightening cycle. Factor identities persist, but loadings shift materially. PC1's 2Y loading rises from 0.27 to 0.42 because the short end, previously anchored near zero, became highly volatile, and PC3's variance share drops from 1.69% to 1.07%. The implication is that a trade hedged to be factor-neutral at inception silently accumulates level and slope exposure as loadings drift.

Factor-Neutral Trade

A 2Y/7Y/30Y butterfly with the belly at 7Y and a long belly base weight of +1 was solved for zero PC1 and PC2 dollar exposure using DV01-weighted loadings, yielding factor-neutral weights of -1.82, +1.00, and -0.23 for 2Y, 7Y, and 30Y legs. The naive equal-DV01-wing butterfly produces weights of -1.60, +1.00, and -0.18, and carries residual exposure of -8.1% in PC1 and positive 2.3% in PC2 relative to its curvature exposure. Naive DV01-neutrality implicitly assumes parallel shifts. Since PC1 is not perfectly parallel and PC2 hits the wings asymmetrically, the naive butterfly runs an unintended level and slope bet.

Strategy & Headline Metrics

The signal is the cumulative PC3 score tracked via a 60-day trailing z-score with no lookahead. The strategy enters when the z-score exceeds 2 in either direction and exits when the absolute z-score falls back below 0.5. Transaction costs are assumed at 0.2bp per leg. Weights and all diagnostics were estimated on in-sample data covering 2019 to 2023, which spans both the low interest rate and hiking regimes; the out-of-sample period covers 2024 to 2025.

The raw cumulative score fails the ADF test in-sample because curvature trended through the hiking cycle rather than reverting, but the demeaned signal the strategy actually trades passes with a half-life of 48 days in-sample and 25.5 days out-of-sample. Mean reversion is real but slow. The backtest delivers a Sharpe of approximately -3.0 in both windows, with annualized gross return nearly zero and transaction costs accounting for virtually all losses. The signal makes no wrong directional bets but lacks enough edge to overcome even minimal transaction costs. The strategy is not production-ready: a 48-day half-life means reversion is too slow to cover transaction costs, stale weights silently accumulate unintended level and slope exposure as regimes shift, and a viable version would need dynamic weight re-estimation, lower execution costs, and a regime filter.